PCB Chronolog

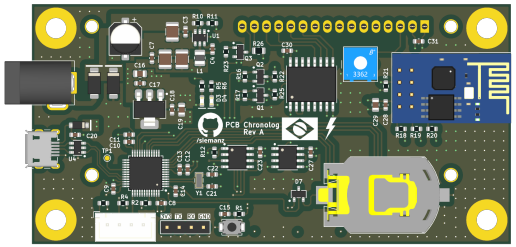
Variant: NO VARIANT

2026-10-15

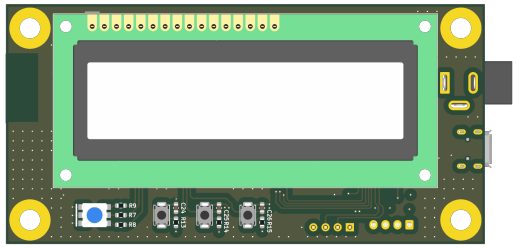
Rev A

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	Assembly Drawing - Bottom	21	.....	31	.....
2	Block Diagram	12	.....	22	.....	32	.....
3	Power Supply	13	.....	23	.....	33	.....
4	MCU	14	.....	24	.....	34	.....
5	RTC and NOR	15	.....	25	.....	35	.....
6	Buttons and Sensors	16	.....	26	.....	36	.....
7	ESP-01	17	.....	27	.....	37	.....
8	LCD	18	.....	28	.....	38	.....
9	Board Layers	19	.....	29	.....	39	.....
10	Assembly Drawing - Top	20	.....	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

Not fitted components are marked as X

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

CHECKED 14-OCT-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

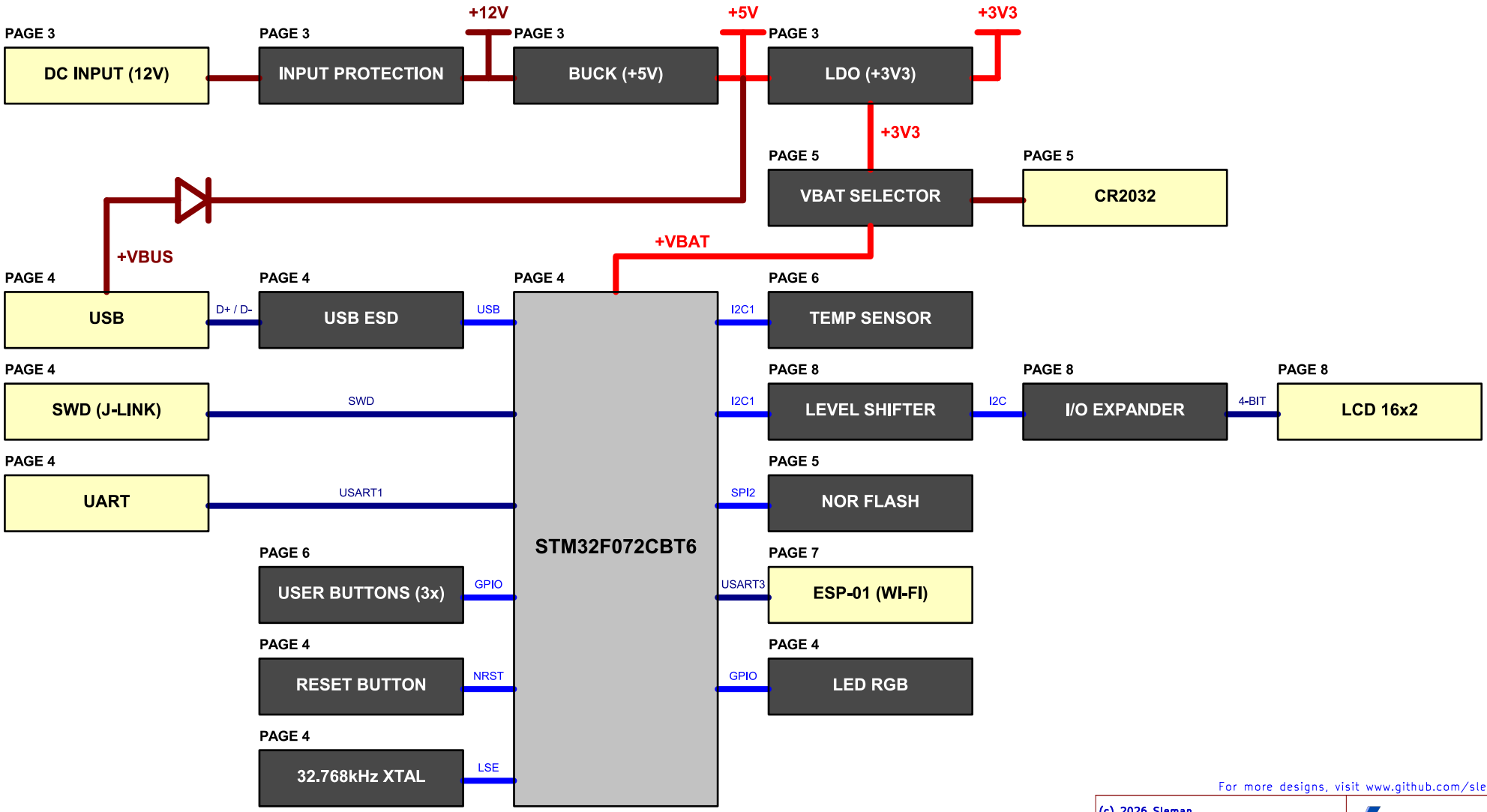
For more designs, visit www.github.com/slemanz

(c) 2026 Sleman

SLEMAN

Title: PCB Chronolog		
Designer: William Sleman	Variant: NO VARIANT	Rev: A
Date: 2026-09-01		Sheet: 1 of 11

[02] - Block Diagram



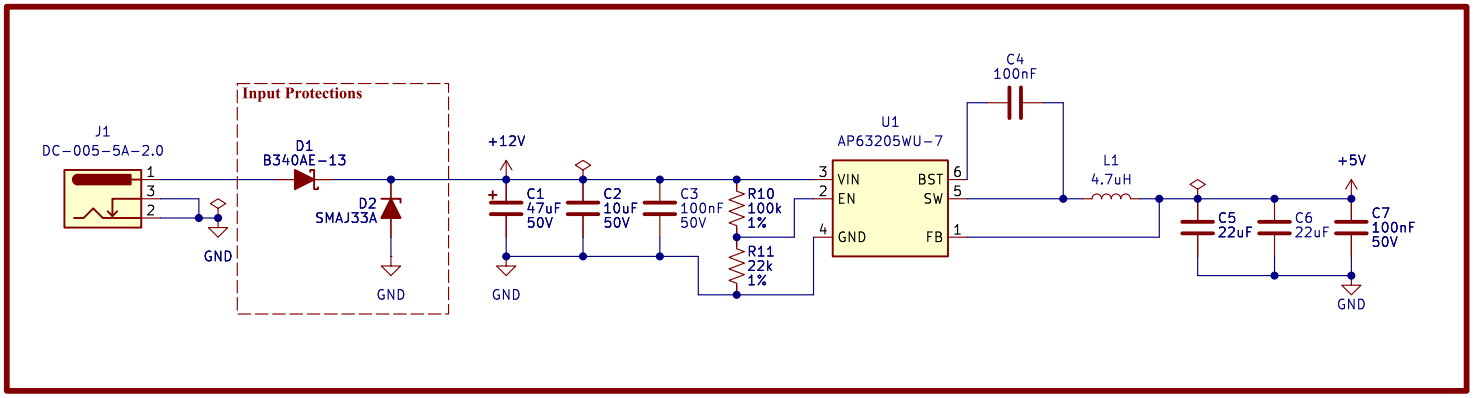
For more designs, visit www.github.com/slemanz

(c) 2026 Sleman

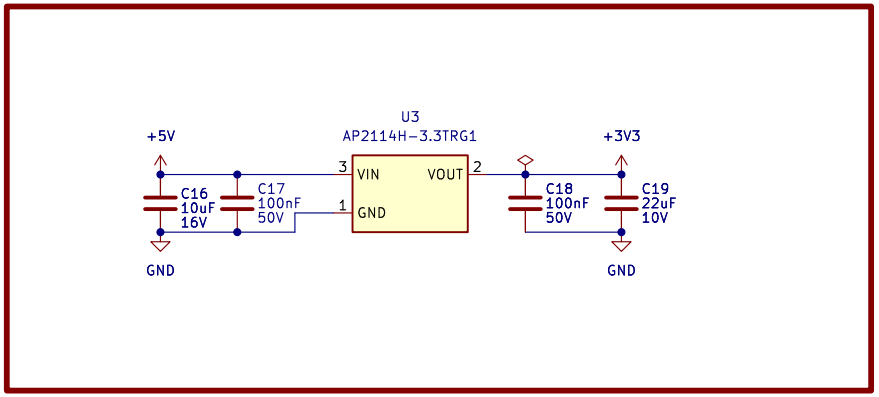
Title: PCB Chronolog		
Designer: William Sleman	Variant: NO VARIANT	Rev: A
Date: 2026-09-01		Sheet: 2 of 11

[03] - Power Supply

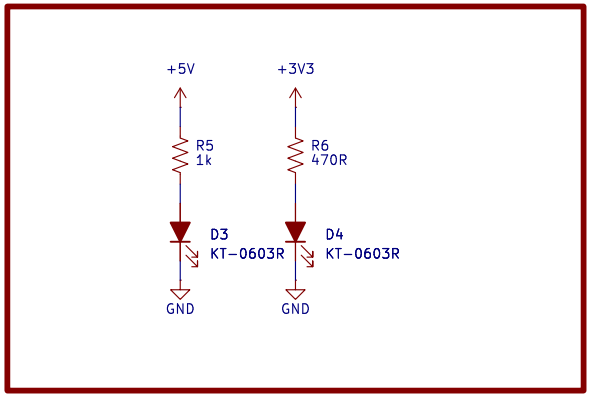
BUCK REGULATOR (+5V)



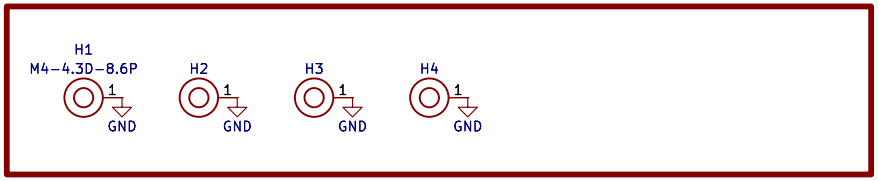
LDO REGULATOR (+3V3)



LED INDICATION



Mechanical

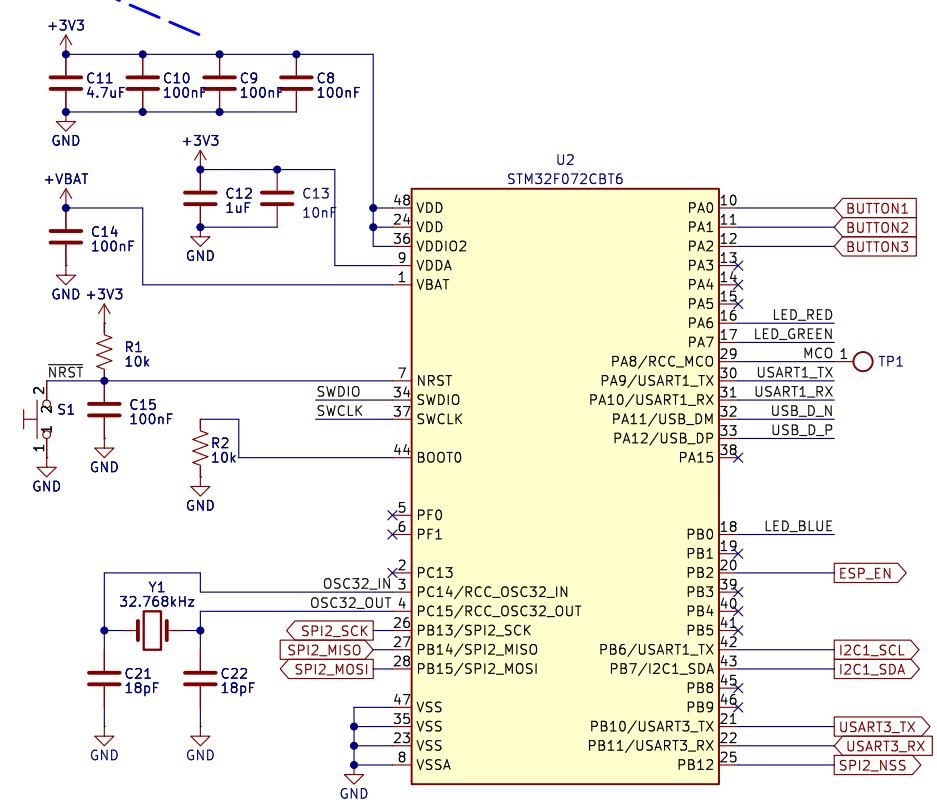


For more designs, visit www.github.com/slemanz

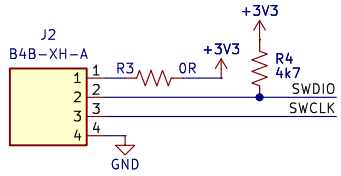
(c) 2026 Sleman		SLEMAN
Title: PCB Chronolog		
Designer: William Sleman	**Variant:** NO VARIANT	**Rev:** A
Date: 2026-09-01		**Sheet:** 3 of 11

[04] - MCU

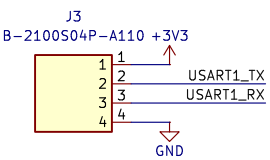
LAYOUT NOTE:
Place each 100nF decoupling capacitor as close as possible to its respective VDD pin.



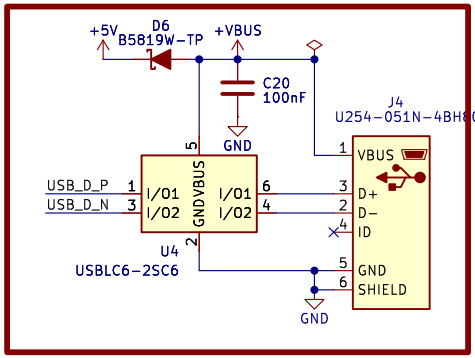
J-LINK CONNECTOR



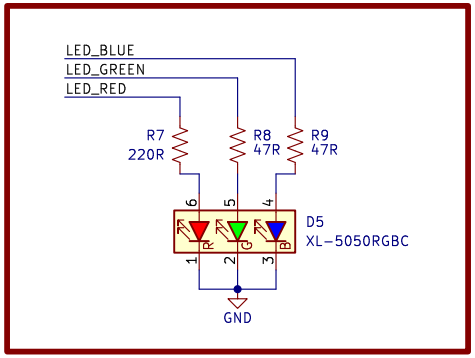
UART CONNECTOR



USB



LED RGB



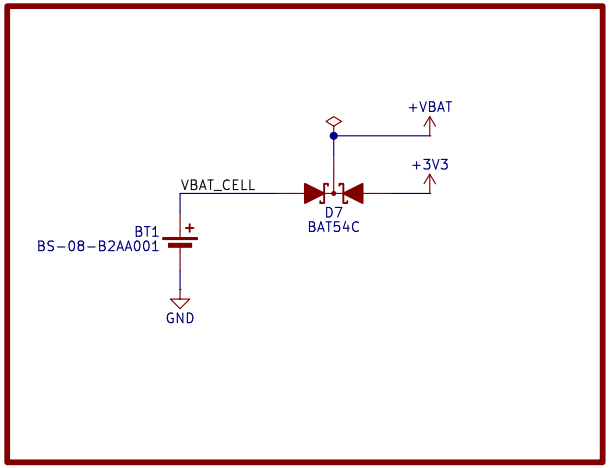
For more designs, visit www.github.com/slemanz

(c) 2026 Sleman

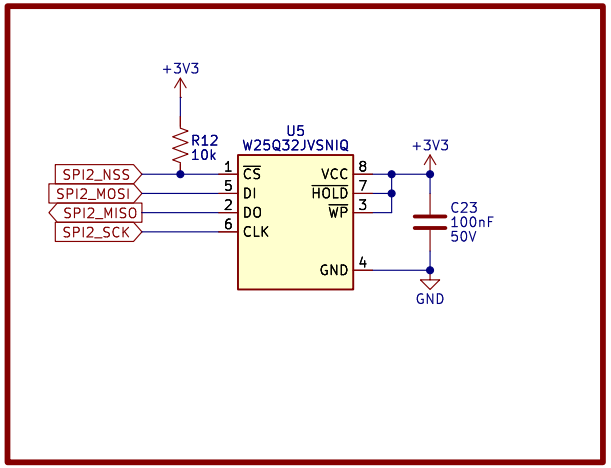
Title: PCB Chronolog		SLEMAN
Designer: William Sleman	**Variant:** NO VARIANT	**Rev:** A
Date: 2026-09-01		**Sheet:** 4 of 11

[05] - RTC and NOR

RTC BATTERY



NOR MEMORY

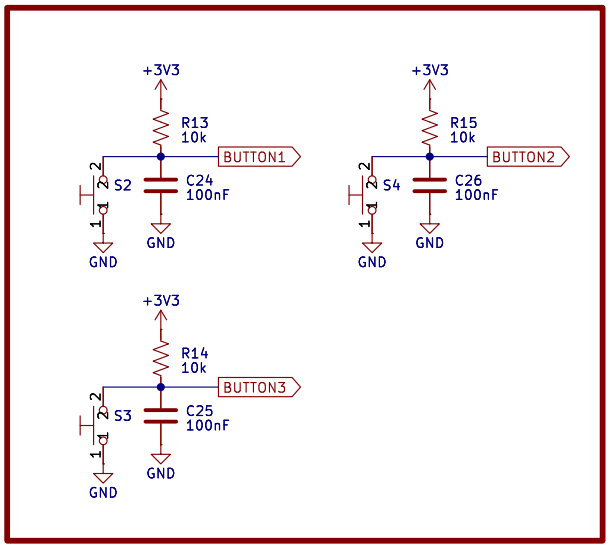


For more designs, visit www.github.com/slemanz

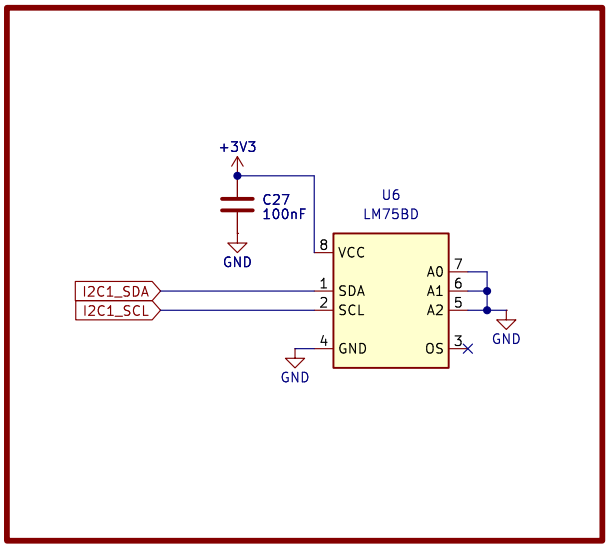
(c) 2026 Sleman		SLEMAN
Title: PCB Chronolog		
Designer: William Sleman	**Variant:** NO VARIANT	**Rev:** A
Date: 2026-09-01		**Sheet:** 5 of 11

[06] - Buttons and Sensors

Buttons



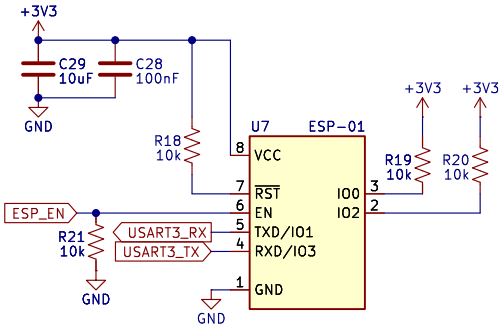
Temperature Sensor



For more designs, visit www.github.com/slemanz

(c) 2026 Sleman		SLEMAN
Title: PCB Chronolog		
Designer: William Sleman	**Variant:** NO VARIANT	**Rev:** A
Date: 2026-09-01		**Sheet:** 6 of 11

[07] - ESP-01



J5
2x4-2.54-TH

For more designs, visit www.github.com/slemanz

(c) 2026 Sleman

SLEMAN

Title: PCB Chronolog

Designer: William Sleman

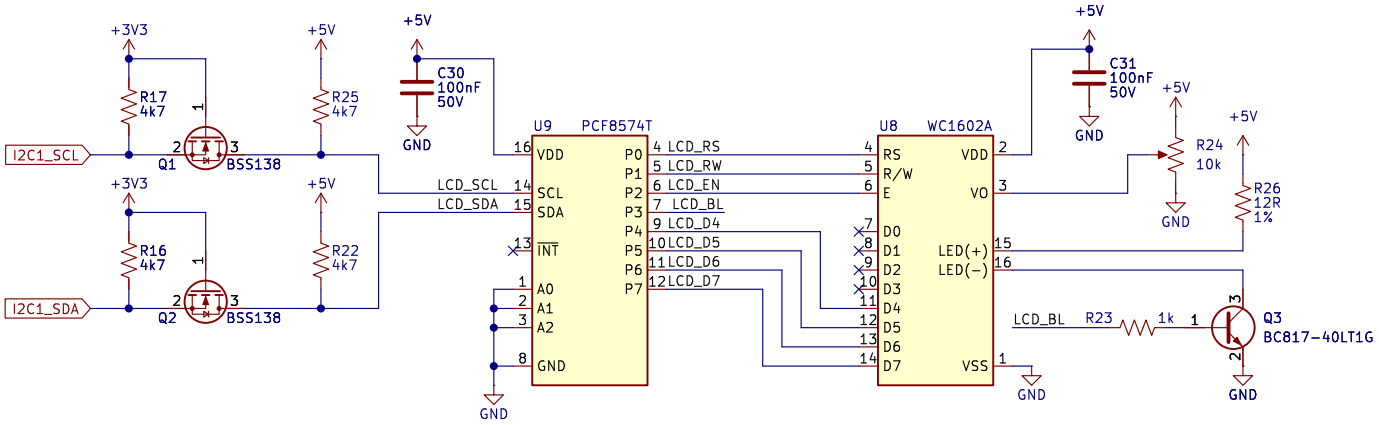
Variant: NO VARIANT

Rev: A

Date: 2026-09-01

Sheet: 7 of 11

[08] - LCD



J6
1x16-2.54-TH

P1
1x16-2.54-TH

For more designs, visit www.github.com/slemanz

(c) 2026 Sleman

SLEMAN

Title: PCB Chronolog

Designer: William Sleman

Variant: NO VARIANT

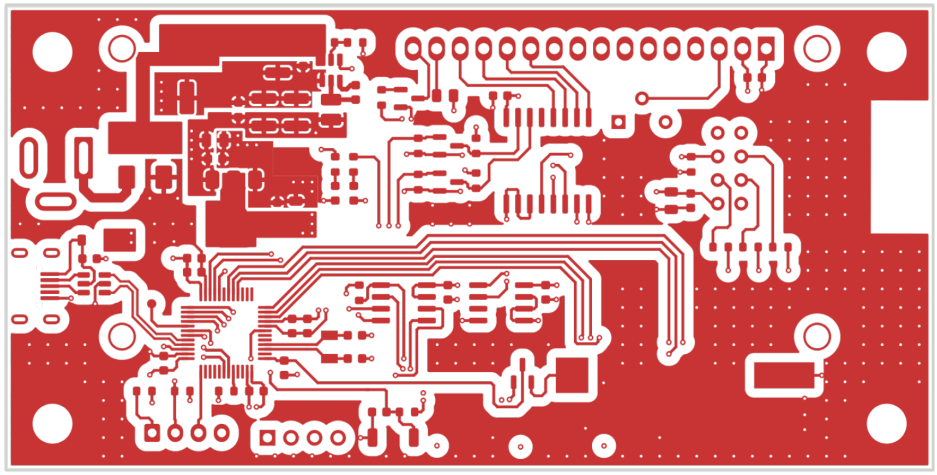
Rev: A

Date: 2026-09-01

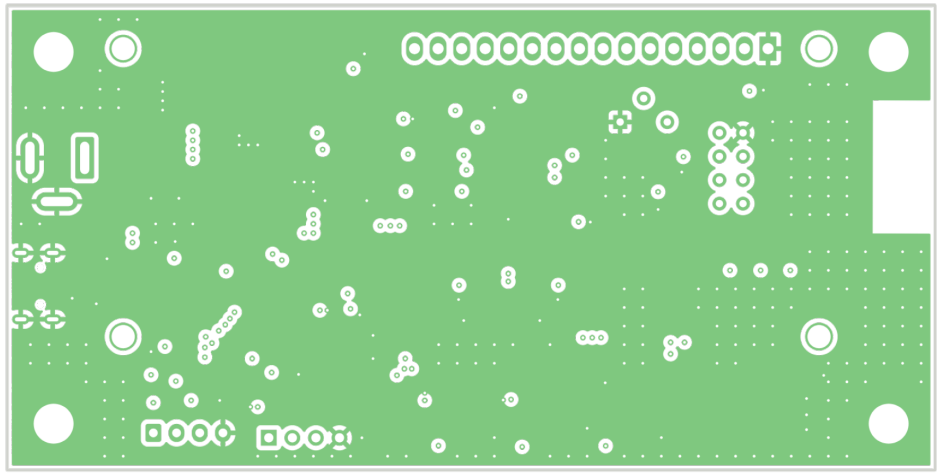
Sheet: 8 of 11

[09] - Board Layers

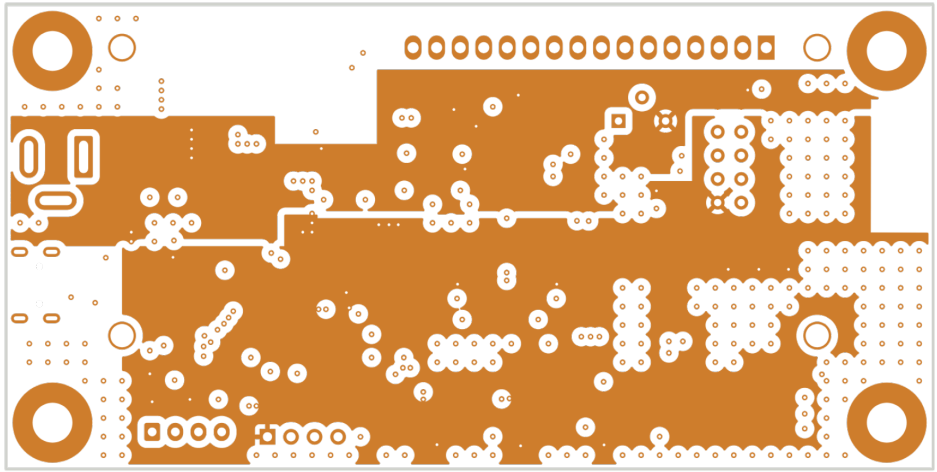
TOP



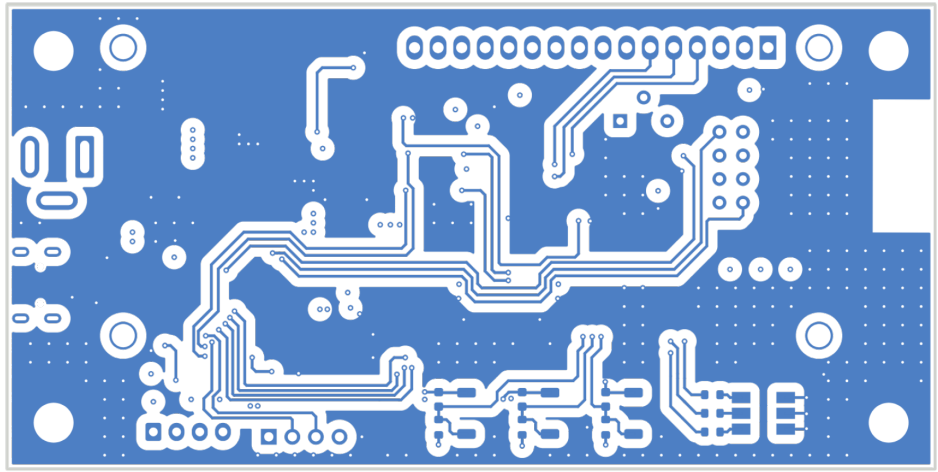
IN1



IN2



BOTTOM

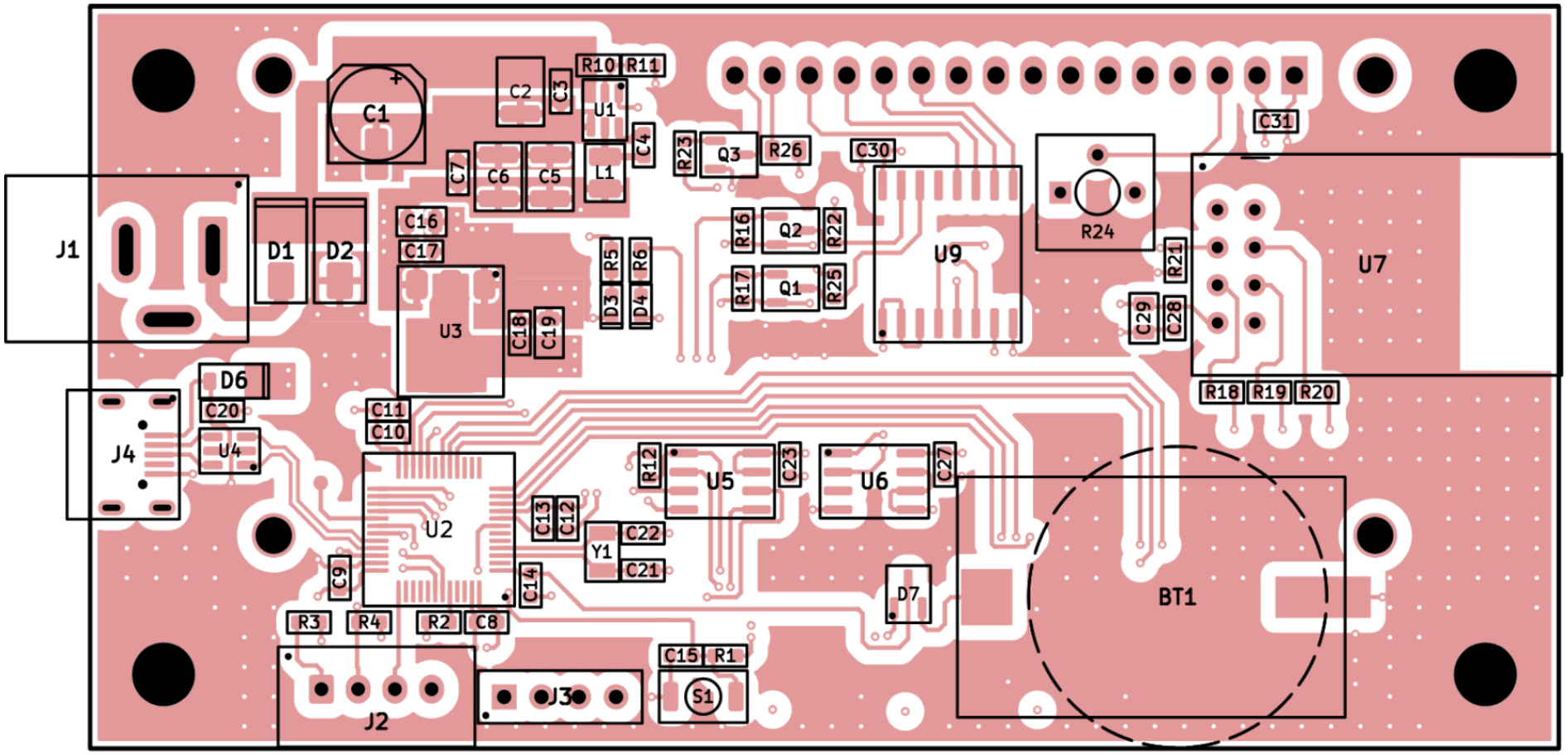


DESIGN NOTE:
All layers are viewed from the top side of the board (not mirrored).

For more designs, visit www.github.com/slemanz

(c) 2026 Sleman		SLEMAN
Title: PCB Chronolog		
Designer: William Sleman	Variant: NO VARIANT	Rev: A
Date: 2026-09-01		Sheet: 9 of 11

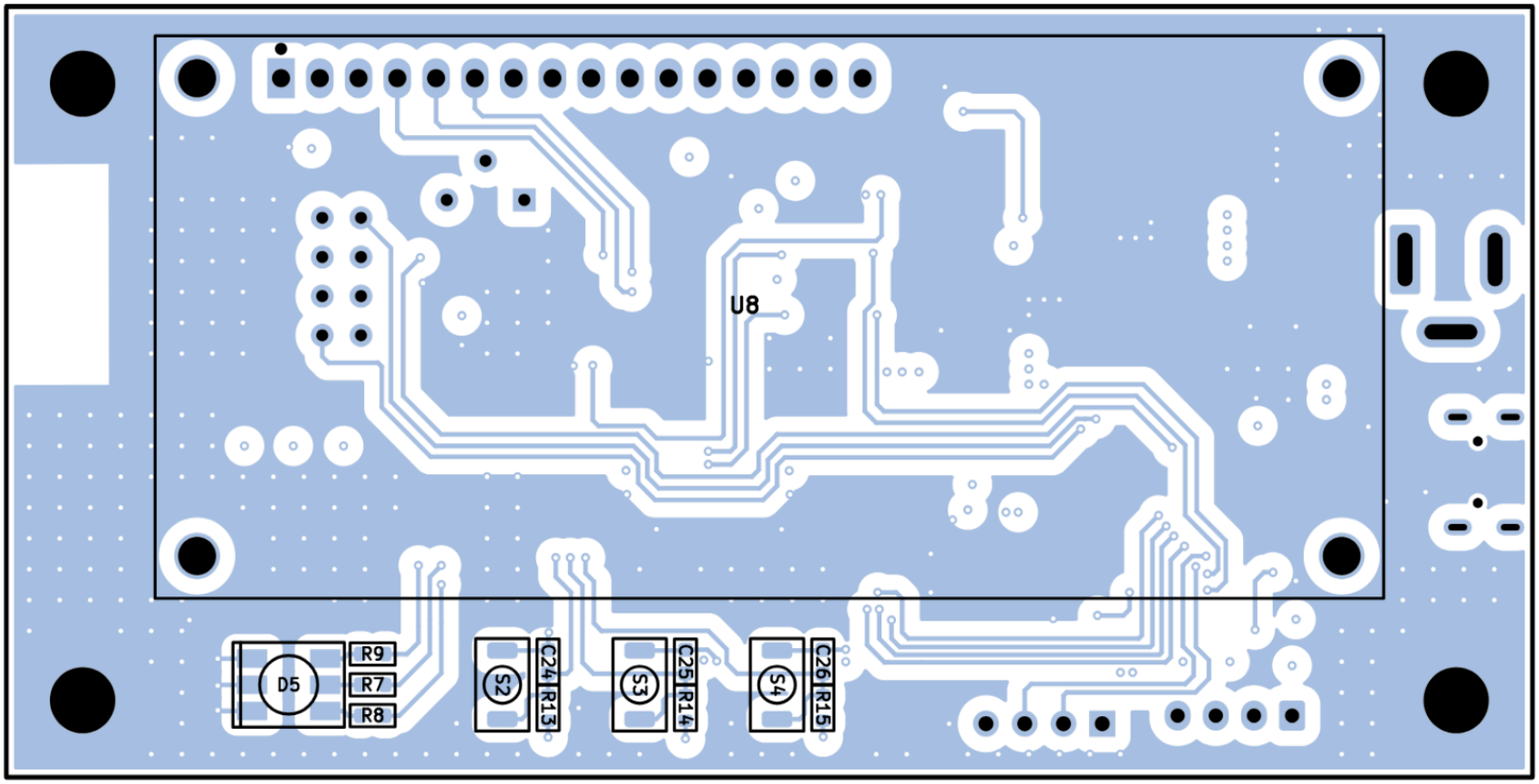
[10] - Assembly Drawing - Top



For more designs, visit www.github.com/slemanz

(c) 2026 Sleman		SLEMAN
Title: PCB Chronolog		
Designer: William Sleman	**Variant:** NO VARIANT	**Rev:** A
Date: 2026-09-01		**Sheet:** 10 of 11

[11] - Assembly Drawing - Bottom



For more designs, visit www.github.com/slemanz

(c) 2026 Sleman		SLEMAN
Title: PCB Chronolog		
Designer: William Sleman	**Variant:** NO VARIANT	**Rev:** A
Date: 2026-09-01		**Sheet:** 11 of 11

SLEMAN

BILL OF MATERIALS

PCB Chronolog

Designed by: William Sleman
Report Date: 10-OCT-2026
Revision: A
Total parts: 90

#	Reference	Qty	Description	MPN	LCSC	Package	Unit Price	Total
1	BT1	1	BATTERY HOLDER COIN CR2032 SMD	BS-08-B2AA001	C964777	SMD		
2	C1	1	CAP ALUM 47uF 20% 35V SMD	VZT470M1HTT-0606L	C46682638	D6.3x5.8		
3	C2	1	CAP CER 10uF 50V X7R 1210	GRM32ER71H106KA12L	C77102	1210		
4	C3, C4, C7, C8, C9, C10, C14, C15, C17, C18, C20, C23, C24, C25, C26, C27, C28, C30, C31	19	CAP CER 100nF 50V X7R 0603	CC0603KRX7R9BB104	C14663	0603		
5	C5, C6	2	CAP CER 22uF 25V X7R 1210	CL32B226KAJNNNE	C309062	1210		
6	C11	1	CAP CER 4.7uF 16V X5R 0603	GRM188R61C475KE11D	C77045	0603		
7	C12	1	CAP CER 1uF 25V X7R 0603	GCM188R71E105KA64D	C85862	0603		
8	C13	1	CAP CER 1nF 50V X7R 0603	0603B103K500NT	C57112	0603		
9	C16, C29	2	CAP CER 10uF 16V X5R 0805	GRM21BR61C106KE15L	C77075	0805		
10	C19	1	CAP CER 22uF 10V X5R 0805	GRM21BR61A226ME51L	C77074	0805		
11	C21, C22	2	CAP CER 18pF 50V C0G 0603	CL10C180JB8NNNC	C1647	0603		
12	D1	1	DIODE SCHOTTKY 40V 3A SMA	B340AE-13	C168859	SMA(DO-214AC)		
13	D2	1	TVS DIODE 33VWM 53.3VC DO214AC	SMAJ33A	C223988	SMA(DO-214AC)		
14	D3, D4	2	LED RED 620-630nm 2V 20mA 0603	KT-0603R	C2286	0603		
15	D5	1	TOP LED 5050 RGB SMD	XL-5050RGBC	C2843868	PLCC-6		
16	D6	1	DIODE SCHOTTKY 40V SOD-123	B5819W-TP	C84354	SOD-123		
17	D7	1	DIODE ARR SCHOTT 30V 200MA SOT23	BAT54C	C2135	SOT-23		
18	J1	1	CONN PWR JACK TH R/A	DC-005-5A-2.0	C381116	TH		
19	J2	1	CONN HEADER JST XH 4POS 2.50mm TH	B4B-XH-A(LF)(SN)	C144395	TH		
20	J3	1	CONN HEADER MALE VERT 4POS 2.54MM	B-2100S04P-A110	C124378	TH		
21	J4	1	CONN RCPT USB MICRO-B 5POS SMD	U254-051N-4BH806	C319170	SMD		
22	J5	1	CONN SOCKET 8POS 2.54mm DUAL ROW TH	PM254-2-04-Z-8.5	C2897405	TH		
23	J6	1	CONN SOCKET 16POS 2.54mm SINGLE ROW TH	KH-2.54FH-1X16P-H8.5	C2905421	TH		
24	L1	1	FIXED IND 4.7uH 1.5A ISAT 2.1A 190mΩ 1008	DFE252012F-4R7M=P2	C668313	1008		

#	Reference	Qty	Description	MPN	LCSC	Package	Unit Price	Total
25	P1	1	CONN HEADER MALE VERT 16POS 2.54MM	B-2100S16P-A110	C49423290	TH		
26	Q1, Q2	2	MOSFET N-CH 50V 220mA 3.5R SOT-23	BSS138	C52895	SOT-23		
27	Q3	1	TRANS NPN 45V 500mA SOT-23	BC817-40LT1G	C94412	SOT-23		
28	R1, R2, R12, R13, R14, R15, R18, R19, R20, R21	10	RES 10kΩ ±1% 100mW 0603	0603WAF1002T5E	C25804	0603		
29	R3	1	RES 0Ω 100mW 0603	0603WAF0000T5E	C21189	0603		
30	R4, R16, R17, R22, R25	5	RES 4.7kΩ ±1% 100mW 0603	0603WAF4701T5E	C23162	0603		
31	R5, R23	2	RES 1kΩ ±1% 100mW 0603	0603WAF1001T5E	C21190	0603		
32	R6	1	RES 470Ω ±1% 100mW 0603	0603WAF4700T5E	C23179	0603		
33	R7	1	RES 220Ω ±1% 100mW 0603	0603WAF2200T5E	C22962	0603		
34	R8, R9	2	RES 47Ω ±1% 100mW 0603	0603WAF470JT5E	C23182	0603		
35	R10	1	RES 100kΩ ±1% 100mW 0603	0603WAF1003T5E	C25803	0603		
36	R11	1	RES 22kΩ ±1% 100mW 0603	0603WAF2202T5E	C31850	0603		
37	R24	1	TRIMMER 10K OHM 0.5W PC PIN TOP	3362P-1-103LF	C58159	TH		
38	R26	1	RES 12Ω ±1% 125mW 0805	0805W8F120JT5E	C17445	0805		
39	S1, S2, S3, S4	4	SWITCH TACTILE SPST SMD	TS-1088-AR02016	C720477	SMD		
40	U1	1	IC REG BUCK 5V 2A 3.8V-32V TSOT-23-6	AP63205WU-7	C2071056	TSOT-23-6		
41	U2	1	IC MCU 32BIT 128KB FLASH 48LQFP	STM32F072CBT6	C81720	LQFP-48		
42	U3	1	IC REG LDO 3.3V 1A 2.5V-6V SOT-223	AP2114H-3.3TRG1	C150716	SOT-223		
43	U4	1	TVS DIODE 5.25VWM 17VC SOT23-6	USBLC6-2SC6	C7519	SOT-23-6		
44	U5	1	IC FLASH 32MBIT SPI/QUAD 8SOIC	W25Q32JVSNIQ	C5355146	SOIC-8		
45	U6	1	SENSOR DIGITAL -55C-125C 8SO	LM75BD,118	C34565	SOIC-8		
46	U7	1	ESP8266 WI-FI IOT MODULE	ESP-01	C503581	MOD		
47	U8	1	LCD CHARACTER DISPLAY MODULE 16x2	WC1602A	C5329588	MOD		
48	U9	1	IC EXPANDER 100KHZ I2C 16SO	PCF8574T	C7605	SOIC-16W		
49	Y1	1	CRYSTAL 32.7680KHZ 12.5PF SMD	Q13FC1350000400	C32346	SMD-3215		
	TOTAL	**90**						**R$ 0,00**

Red text = DNP (Do Not Populate). Parts excluded from the BOM in KiCad are not listed.